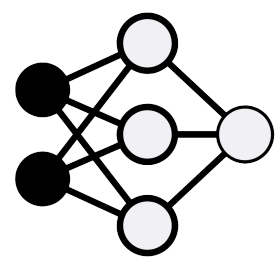


EECS 182/282A | Deep Neural Networks

Spring 2025

Lectures: Mon/Wed 2–3:30 pm, Soda 306



Description

Deep Networks have revolutionized computer vision, language technology, robotics and control. They have a growing impact in many other areas of science and engineering, and increasingly, on commerce and society. They do not however, follow any currently known compact set of theoretical principles. In Yann Lecun's words they require "an interplay between intuitive insights, theoretical modeling, practical implementations, empirical studies, and scientific analyses." This is a fancy way of saying “we don’t understand this stuff nearly well enough, but we have no choice but to muddle through anyway.” This course attempts to cover that ground and show you how to muddle through even as we aspire to do more. That said, we will be leveraging the tentative understanding that we have gained in the past few years.

Lecture reference materials

Lectures are webcast by the department and recordings will be posted to a [youtube playlist](#). You must be logged into your @berkeley.edu account to access the videos. Lectures will have a substantial amount (if not all) of the content covered on the whiteboard (not on presentation slides). **You are expected to take timely handwritten notes (before the next lecture or discussion occurs) and use those to study.**

Because Deep Learning is rapidly evolving field, the material covered in this course can change substantially from semester to semester. If you are interested in materials from previous iterations of this course, please see here: [\[Sp21\]](#) [\[Sp22\]](#) [\[Fa22\]](#) [\[Sp23\]](#)

Syllabus

W	Date	Lecture Topic	Resources	Section	Homework
0	Jan 22	Introduction/Administrivia			HW0 - Basics Written Code
1	Jan 27	Basic Principles			HW1 - Math and Optimization Review Written Code
	Jan 29	Initialization and SGD			
2	Feb 3	Optimization and what is a feature anyway			HW2 - Optimization and Initialization Written Code
	Feb 5	Optimizers			
3	Feb 10	Initializations and start conv-nets			HW3 - CNNs Written Code
	Feb 12	Conv-nets: basics and data augmentations; start normalizations			
4	Feb 17	Holiday			HW4 Written Code
	Feb 19	Finish normalization, dropout, and start Resnet			

W	Date	Lecture Topic	Resources	Section	Homework
5	Feb 24	Finish Resnet and Fully convolutional nets			HW5 Written Code
	Feb 26	Graph Neural Nets			
6	Mar 3	Graph Neural Nets			HW6 Written Code
	Mar 5	Intro to Implementation Issues; Start RNNs			
7	Mar 10	RNNs			HW7 Written Code
	Mar 12	Self-supervision			
8	Mar 17	State-space models			HW8 Written Code
	Mar 19	Attention			
9	Mar 24	Spring Break			
	Mar 26	Spring Break			
10	Mar 31	Transformers			HW9&10 Written Code
	Apr 2	Transformers			
11	Apr 7	Fine-tuning			HW9&10 Written Code
	Apr 9	Prompting			
12	Apr 14	Embeddings, soft-prompting, LoRA			HW11&12 Written Code
	Apr 16	Meta-learning, Transfer Learning			
13	Apr 21	Generative models			HW11&12 Written Code
	Apr 23	Generative models			
14	Apr 28	Diffusion models			
	Apr 30	Buffer			
15	May 5	RRR Week			
	May 7	RRR Week			

EECS 182

Weekly Schedule

Office Hours

Staff

Policies

Assignments

Exams

Grading

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